

Supporting Information

MOSAICS Studio: all-atom natural-move Monte Carlo on modern additive force fields in the browser

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Supporting Information

This file collects the settings, counts and file-level detail a reader needs in order to rerun a job or check a number, and nothing that carries an argument in the main text. Every entry is read from a committed file, and the file is named in each caption.

S1. MOSAICS single-point settings

Table S1 lists the settings that turn a MOSAICS Monte Carlo job into a single-point energy evaluation. The committed inputs are `evidence/1efs/<ff>/mosaics/*.input` and `evidence/protein/<ff>/mosaics/*.input`.

Table S1. The MOSAICS input settings common to every run reported in this work, in the engine’s own syntax, with the two that differ between the two test systems at the foot.

Setting	Value	Effect
\prop_tors_sig	0.0	torsional proposal width
\prop_trans_sig	0.0	translational proposal width
\prop_rot_sig	0.0	rotational proposal width
\total_step_mc	1	one Monte Carlo step
\local_step_md	1	one inner step per move
\statistics_freq	1	report every step
\energy_report	2	print each term and each part of each term
\implicit_solvent	off	vacuum
\ddd	off	no distance-dependent dielectric
\neutralize	off	net charge kept
\inter_list	none	no neighbour list; all non-excluded pairs
\rinter_switch_length	0	no switching function
\rinter_exclude_length	1000	no interaction excluded by distance
\write_energy_unit	kcal	output in kcal/mol
\temperature	300	read, but nothing is propagated
\prop_type	tors (1efs), cart (peptide)	proposal coordinate
\time_step_md	0.4 (1efs), 0.0 (peptide)	zero is required in Cartesian mode

S2. Deck sizes

Table S2 gives the entry counts the engine prints as it reads each deck, taken from the Database reading done lines of `mosaics_sp.out` in each run directory. The counts are properties of the deck and not of the test system: they say how many parameter rows were available, not how many were used, and the engine does not report how many rows a given structure selects.

Table S2. Entries read from each file of each deck, from the mosaics_sp.out of the run named in the last column. The .onfo and .vdw files carry the same number of rows by construction.

Deck	.bond	.bend	.tors_and_impr	.onfo	.vdw	Run
db_ff14sb	90	258	526	595	595	protein/ff14sb
db_ff19sb	102	329	690	666	666	protein/ff19sb
db_bs0	175	567	1060	2145	2145	1efs/bs0
db_bsc1	144	350	660	2211	2211	1efs/bsc1
db_ol15_ol3	145	371	487	1596	1596	1efs/ol15_ol3
db_ol21_ol3	150	388	557	1596	1596	1efs/ol21_ol3

The ff19SB deck carries a sixth file. On the peptide the engine reads 44 .cmap database entries, loads 16 maps at a grid of $N = 24$ and selects four of them, one per interior residue: evidence/protein/ff19sb/mosaics/mosaics_sp.out line 147 reads cmap: 16 map(s) loaded (N=24), 4 backbone term(s). The CHARMM36 nucleic deck is counted differently, because it is generated rather than converted row by row; its manifest, params/db_charmm36_na/manifest.tsv, records 24 residues, 237 bonds, 562 bends, 1036 torsion entries and 15 753 non-bonded pairs.

S3. Reference-engine control files

Table S3 reproduces the three control files in full. Each is committed at the path given in Table 1 of the main text. The GROMACS cutoff is constructed per structure rather than chosen: for a largest interatomic distance d , the cubic box edge is set to $2d + 4$ nm and the cutoff to half of it, $d + 2$ nm, which puts every pair in the molecule inside the cutoff with 2 nm to spare and every periodic image outside it. That gives 5.775263 nm on 1efs and 2.973662 nm on the peptide. The run confirms the cutoff does nothing: sp.log reports Potential shift: LJ r^{-12} : 0.000e+00 r^{-6} : 0.000e+00, Coulomb -0e+00.

Table S3. The complete control settings for the three reference engines. rlist, rcoulomb and rvdw are set equal to each other and per structure, as described above.

Engine	Setting	Value
sander	imin	1
	maxcyc	0
	ntb	0
	cut	999.0
	ntpr	1
	dielc	1.0
OpenMM	nonbondedMethod	NoCutoff
	constraints	None
	rigidWater	False
	removeCMMotion	False
	implicitSolvent	None
	platform	Reference, double precision
GROMACS	integrator	md
	nsteps	0
	cutoff-scheme	Verlet
	verlet-buffer-tolerance	-1
	coulombtype	Cut-off
	vdwtype	Cut-off
	coulomb-modifier	None
	vdw-modifier	None
	epsilon-r	1
	epsilon-rf	1
	DispCorr	no
	constraints	none

S4. The four 1efs tleap presets

Table S4 sets the four committed tleap inputs against each other. The files are `evidence/1efs/<ff>/topology/build_<ff>.in`. Three things differ between them: the parameter files loaded, an `addAtomTypes` block that only the `bsc1` preset carries, and the residue names used for the RNA strand. The last two follow from the libraries. `parmBSC1.lib` uses the carbon types C1, C2, C1 and CE, which `oldff/leaprc.ff99` either does not define or defines with a different hybridisation, so they have to be declared before the library is loaded; the block is the one AmberTools ships in its own `leaprc.DNA.bsc1`. The `ff99` nucleic library names its RNA units RU, RC, RU5 and RC3, while the OL3 RNA library names the same units U, C, U5 and C3. The residues built are the same either way, which `residue_mapping.tsv` confirms residue by residue, every row carrying `base_match True`.

Table S4. The four 1efs tleap presets. All four build the same padded 30-residue system of 942 atoms and are then trimmed to 818 atoms in 26 residues by removing the first and last residue of each strand. `renamed_atoms` is the field of that name in `build_metadata.json`.

Preset	Parameter files loaded	RNA strand written as	renamed_atoms
OL21/OL3	<code>leaprc.DNA.OL21</code> , <code>leaprc.RNA.OL3</code>	U5 U ... C C3	0
OL15/OL3	<code>leaprc.DNA.OL15</code> , <code>leaprc.RNA.OL3</code>	U5 U ... C C3	0
<code>parmbsc1</code>	<code>oldff/leaprc.ff99</code> , then <code>addAtomTypes</code> for C1, C2, C1, CE, then <code>frcmod.parmbsc0</code> , <code>parmBSC1.lib</code> , <code>frcmod.parmbsc1</code>	RU5 RU ... RC RC3	88
<code>parmbsc0</code>	<code>oldff/leaprc.ff99bsc0</code>	RU5 RU ... RC RC3	176

The `renamed_atoms` field counts substitutions performed by `rename_atoms_to_modern` in `tools/build_1efs_amber_reference.py`, which walks the built structure and rewrites seven legacy AMBER atom names into their PDB v3 equivalents: O1P and O2P to OP1 and OP2, H5'1 and H5'2 to H5' and H5'', H2'1 and H2'2 to H2' and H2'', and HO'2 to HO2'. It changes names only, and it runs after tleap has assigned every type and charge, so it cannot alter an energy. The two counts follow from which library supplied each strand. In `bs0` both strands come from `ff99bsc0` and carry the legacy names throughout, giving 2×28 phosphate oxygens on the 28 residues that have a phosphate, 2×30 H5' hydrogens, 30 on the DNA H2' pair and 30 on the RNA H2'/HO'2 pair, which is 176. In `bsc1` the DNA residues come from `parmBSC1.lib`, which already writes OP1, OP2, H5', H5'', H2' and H2'', so only the `ff99` RNA strand is rewritten: 28 phosphate oxygens on its 14 phosphates, 30 H5' hydrogens, 15 H2'1 and 15 HO'2, which is 88. The OL libraries write the modern names on both strands and need none.

S5. Reading a number out of the committed files

Table S5 gives one command per engine, to be run from the `evidence/` folder.

Table S5. Commands that print the committed energies. Paths are relative to `evidence/`.

Engine	Command
sander	<code>awk '/FINAL RESULTS/,0' */sander/*_sander_sp.out</code>
OpenMM	<code>cat */openmm/openmm_sp.out</code>
GROMACS	<code>tail -1 */gromacs/sp.xvg</code>
MOSAICS	<code>grep 'Initial .* energy' */mosaics/mosaics_sp.out</code>

S6. The CHARMM36 conversion route

The full account is `evidence/charmm36.txt`. Four points of detail sit behind the summary in the main text.

The release. Two OpenMM force-field XML files of the CHARMM36 nucleic acid parameters were available, and only one of them can be used. `charmm36.xml` cannot build a strand: its five nucleic templates each declare a single external bond, on O3' and not on P, and none of its 107 patches is a 5' patch, so it

describes mononucleotides and OpenMM cannot produce a reference energy from it. `charmm36_2024.xml` declares external bonds on both O3' and P and ships the 5' patches 5TER, 5MET, 5PHO and 5POM, and is the file used; it carries 100 295 atom types against the 412 of `charmm36.xml`. It also generates its impropers when it loads instead of declaring them in the file, so the deck is read out of the built OpenMM System by `tools/openmm_system_to_mosaics.py` rather than out of the XML, and the reference energies come from `tools/charmm_reference_energy.py` on the same file.

Urey-Bradley. The 1–3 harmonic is written into the bond slot of the `~bend_parm` record, which the engine already evaluates as $\frac{1}{2}k(r_{13} - r_0)^2$. Of 810 unordered triples, 807 are emitted and three are dropped and counted: the P–ON2–P entries and their P3 variants carry a negative force constant, which the deck reader rejects for the whole file. They are pyrophosphate linkages and do not occur in a phosphodiester backbone.

Harmonic impropers. The 188 harmonic impropers are carried as `~torsion_parm` records with `\pot_type{harm}`, which the deck reader expands into a truncated sixth-order Fourier series. That truncation is the only genuine approximation in the conversion, and Table S6 gives its error against the harmonic it replaces. Nucleobase impropers are stiff and stay inside 20° of equilibrium in the structures used here; the series is not exercised further out by any run in this paper.

Table S6. Error of the truncated sixth-order Fourier expansion against the harmonic improper it replaces, as a fraction of the harmonic energy at that displacement from equilibrium.

Displacement from equilibrium	Error
10°	exact to four figures
30°	–0.2 %
45°	–1.8 %
60°	–7.7 %

Generating the deck. A CHARMM36 deck has to be generated from a structure carrying every residue in every position it will be used in, so the committed deck comes from 24 residues covering each of the four DNA and four RNA bases as 5', internal and 3'. MOSAICS builds a torsion for every bonded one-four pair and stops if the deck has no row for it, whereas CHARMM simply omits an unparameterised quadruplet, so the converter writes an explicit zero-energy row in those cases: 197 of them on this deck.

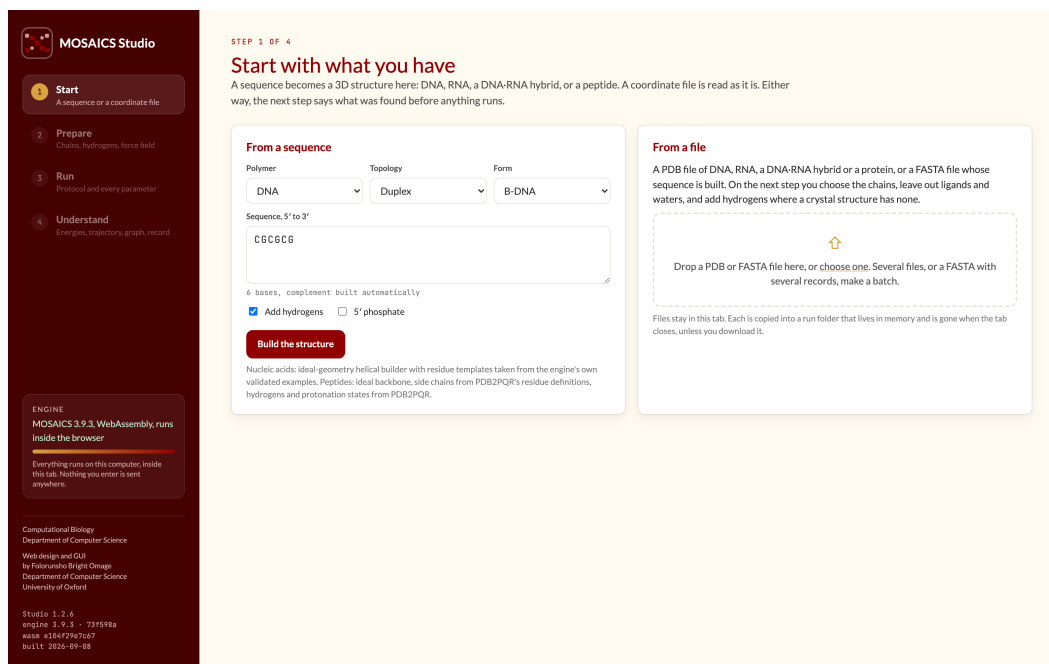
S7. The 1efs coordinate edits

Model 1 of PDB entry 1EFS was taken as deposited and edited in two ways: six atoms were built, and four further operations rename or renumber what was already there. Nothing else about the coordinates was touched – the hydrogens are the deposited NMR hydrogens, none was rebuilt, and no minimisation was performed in this or any other force field. The pipeline is `build_1efs_pipeline.sh` with the four scripts it calls.

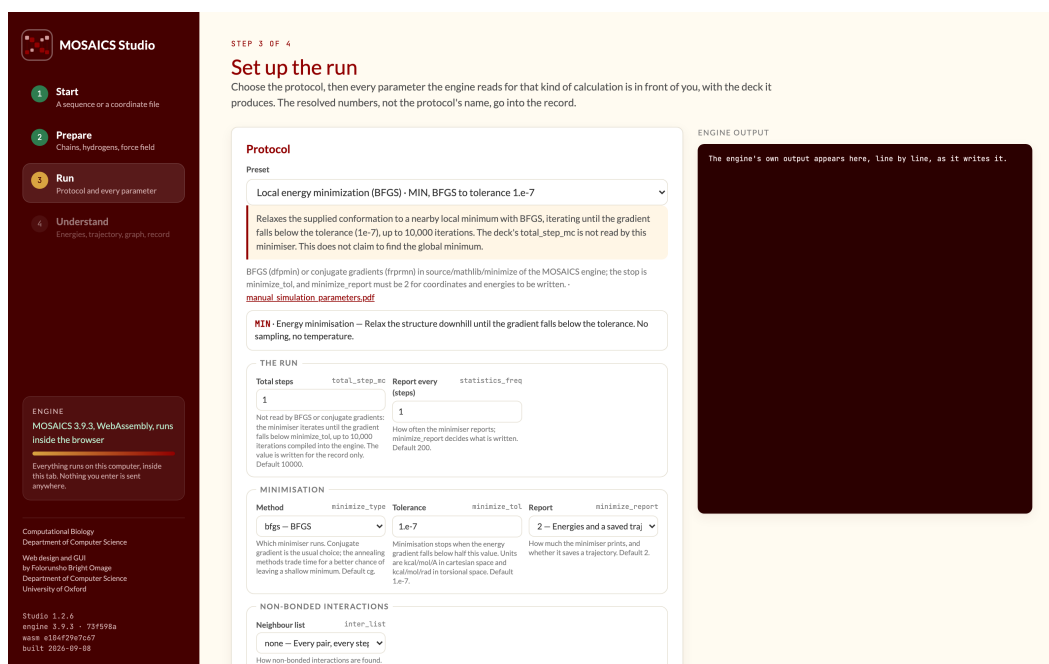
The six built atoms are two 5' phosphate groups, P, OP1 and OP2, one on chain A residue 1 and one on chain B residue 26. Both strands open on a 5' terminus without a phosphate, and the engine's bulk-residue templates expect one. Each group was built by taking the neighbouring residue's phosphate and applying the rigid transformation that maps that neighbour's C5'–O5' frame onto the terminal residue's, so the geometry of the added group is the deposited geometry of its neighbour and no internal coordinate was invented.

The four further edits are: OP1 and OP2 renamed to O1P and O2P; the RNA H2' and HO2' pair swapped to the MOSAICS convention; the 3' HO3' dropped on the two 3' residues; and chain B renumbered so that ascending residue number runs 5' to 3'.

S8. The page, Start and Run steps



(a)



(b)

Figure S1. MOSAICS Studio in the browser, version 1.2.6, the two steps not shown in the main text. (a) Start: a sequence (DNA, RNA or a peptide), a topology and a helical form, or a coordinate file; the default CGCGCG duplex is the one used for the cross-check of Table 8 of the main text. (b) Run: the protocol chosen from the fifteen the Studio offers, here the BFGS local minimisation with its deck and its manual reference, and every parameter the engine reads for that kind of calculation, grouped as the engine's manual groups them; the engine's own output appears line by line in the right-hand panel as it runs.